

Mini-Project 2

1 State-Space Control of the Inverted Pendulum

Using the Simulink model you built in Mini-Project 1, write the system in state-space form about the upright equilibrium and design a state-feedback controller that holds the pendulum upright.

Run two tests. In the first, start the pendulum near upright, perturb it, and let the controller bring it back. In the second, excite the base with a random disturbance and let the controller hold the pendulum upright against it. Plot the pendulum angle $\theta(t)$ for both. In your report, discuss how well the controller worked and where it started to struggle.

Put your Simulink model and supporting content in the appendix.